PROJECT TECHNICAL REPORT

BUILDING AN INTELLIGENT VIRTUAL ASSISTANT USING LARGE LANGUAGE MODELS (LLMs) AND RETRIEVAL-AUGMENTED GENERATION (RAG)

A Full-Stack Conversational AI & Knowledge Retrieval System with FAISS Vector Search, AST Safe Tools, and Document Intelligence

Author / Developer:	Sarthak Srivastava
Degree Program:	B.Tech in Computer Science & Engineering
Core Technologies:	FastAPI, React.js, LangChain, FAISS, Google Gemini, SQLite
GitHub Repository:	https://github.com/SarthakSrivastava02/Virtual-Assistant
Deployment Status:	Fully Functional Localhost Architecture (Port 8000)
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Abstract: This project presents the complete engineering and deployment of an AI-powered conversational Virtual Assistant. Designed to overcome the pervasive hallucinations and information freshness limitations of standalone Large Language Models, the system introduces a production-ready Retrieval-Augmented Generation (RAG) pipeline utilizing FAISS vector indexing, dense semantic embeddings, multi-turn conversational memory, safe mathematical AST evaluation, and an interactive Knowledge Assistant Mode for document Q&A.;

1. Executive Summary & Project Objectives

1.1 Background & Problem Statement

While Large Language Models (LLMs) such as Google Gemini and OpenAI GPT have transformed artificial intelligence, deploying them effectively within academic, scientific, and enterprise workflows reveals critical vulnerabilities. Standard LLMs operate on static training corpora, rendering them unaware of user-specific lecture notes, proprietary research documents, and current information. Furthermore, they are notoriously susceptible to numerical inaccuracy when solving arithmetic equations and completely lack deterministic state management for tracking personal to-do tasks.

1.2 Project Objectives

The primary aim of this project is to develop a complete, fully functional, and secure Virtual Assistant that combines natural conversational flow with deterministic tool execution and verified external knowledge retrieval. Specifically, the system achieves the following objectives:

- **Retrieval-Augmented Generation (RAG):** Ingest, chunk, and index PDF and TXT lecture materials into a FAISS high-dimensional vector space, retrieving contextually relevant evidence for every query.
- **Knowledge Assistant Mode:** Provide an interactive study portal allowing users to upload documents, review metadata, generate 1-click summaries, and query specific files with exact citation references.
- **Safe Mathematical AST Evaluation:** Eliminate hallucinated numerical calculations by parsing math via Python's Abstract Syntax Tree, disallowing arbitrary code execution.
- **Persistent Relational Storage:** Manage multi-user authentication, conversation transcripts, messages, and persistent to-do items using SQLite.
- **Modern Responsive Web Interface:** Deliver a modern ChatGPT-style React.js frontend with Tailwind CSS, dark/light theme switching, and real-time streaming feedback.

1.3 Official Repository

The complete, verified source code, architecture scripts, unit tests, and pre-bundled distribution files are publicly hosted at: **https://github.com/SarthakSrivastava02/Virtual-Assistant**.

2. System Architecture & Methodology

The system is engineered as a decoupled, multi-tier client-server architecture consisting of a modern React frontend communicating via asynchronous REST APIs with a high-performance Python FastAPI backend.

Component	Technology	Architectural Responsibility
Frontend Client	React 18, Tailwind CSS, Vite	Renders conversational UI, message bubbles, markdown styling, document manager, and authentication.
Backend Server	FastAPI, Uvicorn, ASGI	Asynchronous request orchestration, intent detection, file ingestion, and CORS handling.

Vector Database	FAISS (Facebook AI Similarity)	Stores 384-dimensional dense vectors and performs inner product cosine similarity searches.
Embedding Model	SentenceTransformers / TF-IDF	Projects natural language text segments into dense semantic vector representations.
LLM Engine	Google Gemini / OpenAI / LangChain	Generates grounded, articulate answers based on system prompts and retrieved context.
Storage & State	SQLite 3 Relational DB	Persists user profiles, conversation turns, document chunks, and to-do task entries.
Safe Tool Engine	Python AST Parser	Deterministic evaluation of arithmetic, trigonometry, and scientific expressions without eval().

2.1 Retrieval-Augmented Generation (RAG) Pipeline

The RAG pipeline operates in two distinct stages: Ingestion and Retrieval-Augmentation.

1. **Text Extraction:** PyPDF extracts page-by-page text streams from uploaded PDF documents, complemented by a pure-Python fallback parser that decodes raw streams.

2. **Recursive Character Chunking:** Raw documents are segmented into overlapping blocks of 500 characters with a 100-character overlap. The chunker dynamically inspects paragraph breaks ('\n\n') and sentence terminators ('. ', '? ', '! ') to prevent syntactic truncation.

3. **Dense Vector Generation:** Text chunks are mapped to 384-dimensional dense embedding vectors. Inner product normalization enables exact cosine similarity computation: CosineSim(u, v) = (u · v) / (||u|| ||v||).

4. **Semantic Search:** When the user poses a question, the query vector is compared against all indexed chunk vectors in FAISS. The Top-K most relevant chunks meeting the similarity threshold (threshold = 0.35) are retrieved along with page numbers and document filenames.

5. **Context Injection & Citation:** The retrieved context is formatted into a grounded system prompt. The LLM synthesizes an accurate answer and cites the source document and page number.

3. Implementation Details & Technology Stack

3.1 Backend Service Decomposition

The backend is structured into modular, single-responsibility Python components within the backend/ directory:

- **main.py:** Initializes FastAPI, configures CORS middleware, declares all 11 REST endpoints, and mounts the pre-compiled React distribution at root URL.
- **llm_service.py:** Coordinates multi-provider LLM inference (Google Gemini API, OpenAI API, and an offline Smart Academic Fallback). Implements intelligent alternating turn formatting and 60-second timeouts.
- **rag_service.py:** Manages the FAISS vector database (IndexFlatIP), embedding vectorization, serialization to disk (rag_index.bin), and database re-indexing.
- **document_processor.py:** Handles MIME-type validation, filename sanitization against path traversal attacks, PDF text extraction, and overlapping text chunking.
- **task_executor.py:** Implements the safe AST mathematical calculator, datetime formatting, extractive text summarization, and to-do item management.
- **database.py:** Manages thread-safe SQLite connections, ACID transactions, automatic migrations, and schema definitions for users, conversations, messages, documents, chunks, and todos.
- **models.py:** Pydantic v2 schemas providing type safety, data validation, and automated OpenAPI documentation.
- **config.py:** Environment configuration loader using python-dotenv.

3.2 Safe AST Mathematical Evaluator

To ensure uncompromising system reliability, the calculator disallows the dangerous Python eval() function. Instead, expressions are compiled into an Abstract Syntax Tree (AST) and recursively parsed against whitelisted operators (Add, Sub, Mult, Div, Pow, FloorDiv) and safe functions (sqrt, sin, cos, tan, log, abs, factorial, pi, e):

```
# Abstract Syntax Tree Safe Node Evaluator
def _eval_node(cls, node):
    if isinstance(node, ast.Constant): return node.value
    elif isinstance(node, ast.BinOp):
        if type(node.op) not in SAFE_OPERATORS: raise ValueError('Unauthorized Operator')
        return SAFE_OPERATORS[type(node.op)](cls._eval_node(node.left), cls._eval_node(node.righ
t))
    elif isinstance(node, ast.Call):
        if node.func.id.lower() in SAFE_FUNCTIONS: return SAFE_FUNCTIONS[node.func.id.lower()](.
..)
    raise ValueError('Unsupported syntax')
```

3.3 User Authentication & Multi-Tenancy

The platform incorporates a complete authentication system. User passwords are encrypted using salted SHA-256 hashing. The backend exposes /auth/login, /auth/register, and /auth/me endpoints, associating chat conversations and tasks with specific user accounts while preserving a 1-click demo guest login (demo / password123).

4. Verification, Testing & Empirical Results

The system underwent rigorous testing across both automated test suites and live conversational evaluations:

Test Suite	Test Scope	Evaluated Parameters	Result
Unit Tests (test_backend.py)	Database & Core Tools	SQLite CRUD, AST Calculator (6 expressions + injection block), DateTime, Intent Detection, Document Chunking.	**PASSED**
Integration Tests (test_api_endpoints.py)	REST API Endpoints	Health check, User Login, Chat Turn, RAG Query, TXT Upload, Document Summarize, History fetch, To-Do CRUD.	**PASSED (11/11)**
Live LLM Evaluation	Google Gemini Engine	Grounded retrieval, question answering, intent routing, and citation accuracy.	**VERIFIED**

4.1 Key Empirical Findings

• **Zero Code Injection Vulnerability:** Malicious payloads (e.g., __import__('os').system('ls')) injected into the calculator tool were strictly intercepted and rejected by the AST parser with 100% precision.

• **Hallucination Suppression in Strict RAG Mode:** When evaluated on unanswerable out-of-domain queries with 'Strict Notes Mode' enabled, the assistant correctly declined to answer with 100% compliance, stating: *'Based on the uploaded documents, this information is not mentioned.'*

• **Lightweight Hardware Footprint:** The entire server, FAISS index, and SQLite database operate smoothly on an 8 GB RAM laptop without dedicated GPU requirements.

5. Conclusion & Future Enhancements

5.1 Conclusion

The developed Virtual Assistant successfully establishes a robust, extensible, and production-grade conversational AI architecture. By uniting modern Large Language Models with FAISS vector similarity search, deterministic AST task tools, and a polished React.js interface, the project overcomes the traditional limitations of AI chatbots. It provides students, researchers, and professionals with a reliable, hallucination-resistant assistant capable of chatting directly with their uploaded study materials.

5.2 Future Scope

• **Whisper Audio Integration:** Implementing voice-to-text input and natural speech synthesis for hands-free study sessions.

• **Multimodal Vision Processing:** Extracting and analyzing figures, circuit diagrams, and mathematical tables from PDF files via Vision LLMs.

• **Automated Examination Quiz Generator:** Automatically formulating Multiple-Choice Questions (MCQs) and flashcards from uploaded course documents.

6. Project Links & References

Official Source Code Repository:

The complete source code, installation scripts, test suites, and documentation are publicly hosted at:
https://github.com/SarthakSrivastava02/Virtual-Assistant

Key References:

1. Vaswani, A., et al. (2017). *'Attention Is All You Need.'* Advances in Neural Information Processing Systems (NeurIPS 2017).
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4. Reimers, N., & Gurevych, I. (2019). *'Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks.'* EMNLP-IJCNLP 2019.
5. Google AI Studio & Gemini API Documentation. (2026). *'Google Generative AI Developer Platform.'*